

Eduardo Guerrero

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PROFESSIONAL SUMMARY

Electromechanical Engineer with hands-on experience in robotics, autonomous navigation, embedded systems, FPGA design, & feedback control systems. Skilled in ROS2, Python, C/C++, sensor fusion, & autonomous vehicle development. Experienced designing, testing, & deploying hardware-software solutions using NVIDIA Jetson, STM32, MATLAB, & FPGA platforms.

SKILLS

Robotics and Autonomous Systems: ROS 2 (Humble), Nav2, Robot Localization, Sensor Fusion, Coordinate Frame Transformations

Test Equipment: Oscilloscope, Function Generator, Digital Multimeter, Spectrum Analyzer, Network Analyzers

Hardware and Embedded: STM32, ESP32, Microchip MCUs, PLC Programming, Ladder Logic, NVIDIA Jetson Orin (ARM64), FPGA Development (Zynq-7000), Industrial Electronic Troubleshooting

Programming and Software: C, C++, Python, MATLAB, Verilog, Linux, AutoCAD, Fusion360, Microsoft Office, Onshape, KiCAD, LabVIEW, NI Multisim, Cadence

Control and Signal Processing: PID Control Systems, Digital Signal Processing (DSP)

Tools: Protoboard Prototyping, Fine Pitch SMT Soldering, Stencil Use, Benchtop Power Supplies, Logic Analyzers, Wire Harnessing, Digital Calipers

Languages: English (Fluent), Spanish (Fluent)

PROJECTS

Autonomous Mobile Robot: Sensor Fusion and Navigation (ROS 2)

- Engineered an autonomous navigation stack using ROS2 & Nav2, for a static 20x20 meter outdoor map by fusing GPS, IMU and wheel odometry data
- Developed custom Python nodes that records routes and allows for autonomous replay, deployed on an NVIDIA Jetson Orin
- Tuned Extended Kalman Filter parameters for sensor fusion, significantly reducing localization drift during outdoor operation.
- Developed modular nodes & conducted field testing to validate autonomous navigation performance and system reliability.

Feedback Control System: Balance Beam (Arduino Uno)

- Designed a closed-loop PID control system with ultrasonic sensor feedback for real-time position control, iteratively optimizing parameters to minimize steady-state error and overshoot.
- Modeled system dynamics and evaluated transient response characteristics by analyzing position and velocity behavior to refine rise and settling time.

FPGA-Based Game System (Zybo Z7, Zynq-7000)

- Designed and verified a multi-state game controller in Verilog HDL using Vivado targeting the Zynq - 7000 FPGA architecture
- Performed timing analysis and constraint optimization to achieve artifact-free VGA output and deterministic game operation.
- Programmed a light sensor integrated with the core game logic to register user feedback and drive interactive gameplay

Adaptive Noise Cancellation (MATLAB/Simulink)

- Implemented a Normalized Least Mean Squares adaptive filter in MATLAB/Simulink and evaluated convergence characteristics under varying noise conditions.
- Analyzed filter convergence and validated Gaussian noise reduction in simulation

JOB EXPERIENCE

Engineering Intern (June 2024 - August 2024)

The Holt Group

El Centro, CA

- Coordinated distribution of engineering drawings, change orders, and inspection documentation for construction projects, ensuring accurate implementation of design revisions across multiple sites.
- Performed field inspections and site assessments, documenting construction progress, discrepancies, and design compliance through detailed reports and photographic evidence
- Surveyed site infrastructure and generated field sketches supporting project documentation and engineering evaluations.

Aquatics Supervisor of Programs (Jan 2023 - present)

Associated Students Incorporated, Cal Poly Pomona

Pomona, CA

- Managed program budgets and optimized resource allocation to support operational efficiency and program growth.
- Supervised a team of 8 instructors serving 1,000 participants per semester

EDUCATION

B.S., Electronic Systems Engineering Technology – Cal Poly Pomona

A.S., Science for Transfer with Distinction in Engineering - Imperial Valley College